Pose Estimation and Motion Retargeting for Virtual Reconstruction

Yunhuan Wu, Carter Weaver, Xilin Zhang, Qiushi Zhang

***Abstract*— In robotics control and virtual reconstruction, accurate human motion analysis is crucial for intuitive interaction and realistic digital representations. This paper proposes a system that integrates state-of-the-art pose estimation with advanced motion retargeting to bridge the gap between raw video data and adaptable motion models. Our hybrid system enhances pose estimation using OpenPose and MotioNet for pose estimation and motion retargeting based on ImitationNet, ensuring high fidelity in the transfer of human movement to robotic actions. We evaluate our system against existing baselines and demonstrate its effectiveness in a simulated environment, showcasing its potential for applications in robotics control and virtual reconstruction. Our pose estimation component achieves near-zero temporal inconsistency (0.0468) and average limb length variation (5.71%), closely matching ground truth motion capture. Our motion retargeting component outperforms the most commonly used methods, such as the Auto-Rig-Pro (ARP) Blender add-on, with a self-penetration rate of 28.85% on cross-structural retargeting. Our contributions include a robust pose estimation and motion retargeting framework, enhanced preservation of motion dynamics, and comprehensive evaluation in simulated scenarios.**

I. Introduction

In robotics control and virtual reconstruction, accurate human motion analysis is essential for intuitive interaction and realistic digital representations. In this work, we propose a system that integrates state-of-the-art pose estimation with advanced motion retargeting to bridge the gap between raw video data and adaptable motion models for diverse applications.

Pose estimation and motion retargeting are fundamental to robot imitation learning from human demonstrations and online videos. By leveraging methods such as OpenPose [1] and MotioNet [2], one can extract detailed skeletal data from video sequences with high precision. This capability is crucial for robots to accurately mimic human movements, as it ensures that even subtle nuances in human actions are captured. For instance, in a manufacturing setting where a robot needs to learn a complex assembly task from a human demonstrator, precise pose estimation enables the robot to replicate the exact hand and body movements required for the task.

Motion retargeting techniques, such as those used in ImitationNet [3], allow for the adaptation of human movements to the robot's mechanical configuration, ensuring that the robot can perform natural and contextually appropriate motions. This is particularly important in scenarios where the robot's anatomy differs significantly from that of a human. For example, a humanoid robot with a different joint structure can still perform natural and contextually appropriate motions by leveraging our advanced motion retargeting technique.

Fig. 1: 3D Motion Reconstruction (from MotioNet) [2]. The system takes RGB video input (left) and reconstructs a corresponding 3D pose sequence (right), enabling motion understanding from visual observations.

Our system builds upon these advancements by developing a hybrid approach that couples robust pose estimation with advanced motion retargeting. This integration not only enhances the accuracy of motion capture but also ensures adaptability across diverse robotic platforms. For example, ImitationNet demonstrates the ability to control a robot in real-time using human poses estimated from RGB videos [3]. Similarly, DexMV showcases a pipeline for converting human hand-object poses from videos into robot demonstrations via motion retargeting [4], significantly improving robot learning in complex dexterous manipulation tasks.

Furthermore, recent works such as VIEW: Visual Imitation Learning with Waypoints [5] highlight the importance of extracting critical waypoints from human demonstrations to guide robot learning. This approach focuses on identifying key frames and trajectories that capture the critical parts of a task, such as grasping an object, and using these waypoints to iteratively improve the robot's performance. This method is particularly effective in addressing the morphological differences between humans and robots, ensuring that the robot can successfully imitate the demonstrated task.

Prior work in pose estimation has seen impressive strides with the models described in the Related Works section. However, challenges such as occlusion and varying lighting remain, as well as a lack of end-to-end pipelines directly turning video into animation. Similarly, motion retargeting techniques, including those based on the Skeleton Aware Network [6], have advanced the field by offering methods that are more adaptive to different skeleton topologies. Yet, these methods often struggle with maintaining the subtle dynamics of motion, particularly when transitioning between highly distinct skeletal structures. These limitations can impede the overall effectiveness of human-robot interaction

and the realism of virtual reconstructions.

In response to these challenges, our work makes the following contributions:

- We develop a hybrid system that couples robust pose estimation with advanced motion retargeting to ensure high fidelity in the transfer of human movement. This system is especially beneficial for robot imitation learning from human demonstration and online videos, as it provides accurate and adaptable motion data.
- We enhance the baseline retargeting method to better preserve motion dynamics across diverse skeletal configurations, addressing key limitations in current approaches. This enhancement is crucial for robots to learn complex tasks from human demonstrations, ensuring that the subtle nuances of human movements are retained.
- We design a comprehensive evaluation pipeline that examines both pose estimation and motion retargeting from multiple perspectives, including temporal consistency, anatomical correctness, motion smoothness, and surface integrity. This evaluation setup allows us to rigorously assess the generalizability of our method across different skeletons and motion types.

By integrating these components, our approach sets a new benchmark for systems that require both precision and adaptability in translating human motion into robotic and virtual domains.

II. Related Work

A. Pose Estimation

Monocular human pose estimation has evolved significantly over the past decade, with approaches ranging from direct regression methods to complex multi-stage pipelines. Early work in pose estimation relied on handcrafted features along with kinematic models. The field evolved significantly with the introduction of deep learning approaches, which eliminated the need for explicit part detectors or graphical models.

1) CNN-Based Pose Estimation: The work of Google's Toshev and Szegedy (DeepPose) pioneered a shift toward deep learning by formulating pose estimation as a DNN-based regression problem, where joint coordinates are directly predicted from full images using convolutional neural networks [7]. At the same time, the field eventually shifted toward two-stage pipelines where 2D pose estimation is followed by lifting to 3D space. One such pipeline is OpenPose, developed at Carnegie Mellon University [8]. OpenPose is mostly used for 2D estimation. It involves CNN layers for feature extraction and a bottom-up approach using Part Affinity Fields (PAFs) to simultaneously detect every body part while associating each part to a person. For 3D estimation, it requires multiple cameras. However, CNN-based methods like DeepPose may struggle with occlusion and require large amounts of annotated data, and OpenPose, while effective for 2D, lacks native support for 3D pose estimation without multi-view input.

2) Transformer-Based Pose Estimation: Recent advancements have focused on incorporating temporal information for more accurate and smooth pose estimation. MotionBERT, introduced in 2022, is a framework that achieved this goal, using a dual-stream spatio-temporal transformer (DSTformer) to capture long-range relationships among skeletal joints [9]. Just last year, MotionAGFormer innovated on this dual-stream transformer architecture by introducing a graph convolutional network (GCN) on one of the streams [10]. GCNs have proven effective at capturing local relationships between adjacent joints and contribute to MotionAGFormer's state-of the-art results of Human3.6M data. Transformer-based methods such as MotionBERT and MotionAGFormer achieve smoother and more accurate predictions, especially in dynamic settings, but they are often computationally intensive and require large datasets to generalize well.

3) Kinematics-Aware Pose Estimation: Most pose estimation methods focus on joint positions rather than rotations, requiring post-processing for virtual reconstruction applications. However, certain models directly encode kinematic information, allowing for easy conversion to animation. A notable example is VIBE, an SMPL-based pipeline (Skinned Multi-Person Linear Model) that learns to regress SMPL parameters, implicitly encoding pose, shape, and translation for each video frame [11]. As a result, the output of VIBE can be directly converted to a BVH file, which is the input needed for motion retargeting. Another kinematics-aware pipeline that is convenient for motion retargeting is the one being deployed for this project, the details of which are covered in Methods. Kinematics-aware models like VIBE are well-suited for downstream tasks like animation and retargeting due to their structured output, but they tend to be less accurate in fine-grained joint localization compared to coordinate-based methods.

B. Motion Retargeting:

Motion retargeting focuses on transferring motion from a source skeleton to a target skeleton while maintaining motion semantics and realism.

1) Kinematic-Based Motion Retargeting: Traditional optimization-based approaches primarily rely on kinematic constraints and inverse kinematics (IK) to adjust motions [12], [13]. Later methods incorporated additional constraints, such as dynamics constraints [14] and trajectory constraints [15]. However, these methods require hand-crafted constraints and paired motion data, limiting their adaptability.

2) Data-Driven Motion Retargeting: With the advancement of deep learning, data-driven motion retargeting methods have gained increasing attention. Neural Kinematic Network (NKN) [16] leveraged a forward kinematics layer with cycle consistency loss to achieve unsupervised motion retargeting, though it assumes structural similarity between skeletons. PMnet [17] improved unpaired motion retargeting by separately modeling frame-wise poses and global motion trajectories, enhancing temporal consistency and generalization to unseen skeletons. Following this, Skeleton-Aware Network (SAN) [18] proposed a disentangled representation

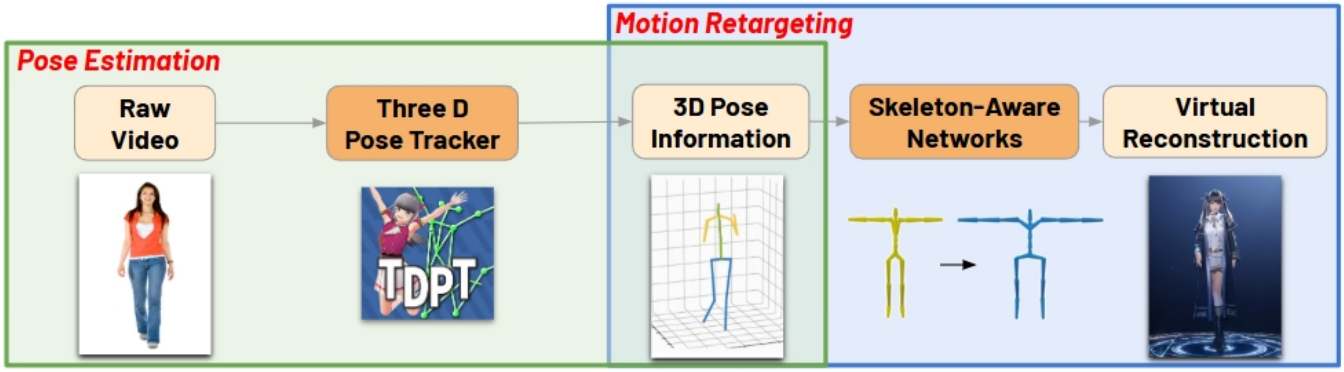


Fig. 2: Virtual Reconstruction Pipeline. 3D information is extracted from the 2D video input in the first place and then transferred between skeletons to achieve accurate and smooth virtual reconstruction.

of motion, skeleton, and style, allowing retargeting across characters with different skeletal topologies.

3) Skin-Aware Motion Retargeting: Beyond articulated skeletons, recent works have considered the interaction between motion and character shape. Residual Retargeting Network [19] introduced a framework for skinned motion retargeting, explicitly modeling character shape and self-contacts to improve motion realism. However, due to complex model and insufficient training data, R2ET performs poorly in terms of computational efficiency and scalability.

III. Methods

We build a pipeline consisting of two major parts: Pose Estimation and Motion Retargeting, illustrated in Figure 2.

A. Pose Estimation

1) ThreeDPoseTracker: For our human pose estimation method, we have adopted ThreeDPoseTracker (TDPT) as our primary solution, based on its superior performance in both accuracy and downstream motion retargeting. TDPT is a real-time 3D pose tracking app that leverages deep learning and computer vision to estimate full-body joint positions and rotations directly from monocular video or live camera input.

The system uses a proprietary neural network architecture trained on large-scale motion datasets to robustly infer 3D joint coordinates and orientations, even under challenging conditions like occlusions or quick movements. TDPT outputs motion data in standard formats including BVH, enabling us to integrate it easily with our motion retargeting pipeline. Below, we outline our core architecture and workflow with TDPT:

- **Inputs:** Accepts monocular RGB video or live webcam input; can also process pre-recorded video files.
- **Neural Network:** Utilizes a deep neural network trained on large-scale motion capture datasets.
  - Detects and tracks 5 human body joints in real time.
  - Employs temporal modeling with optical flow to ensure smooth pose predictions across frames.
  - Uses spatial and kinematic priors to resolve depth ambiguities and maintain anatomically plausible joint rotations.
- **Outputs:** Exports motion data in standard BVH format capturing 3D joint positions and global/local joint rotations.

As discussed further in our Results and Discussions sections, we mainly chose TDPT because it vastly outperforms open source alternatives we investigated, namely OpenPose and Motionet. Using TDPT ensures that our pipeline produces reliable, animation-ready motion data, which minimizes retargeting errors and imrpoves the realism of our virtual reconstructions.

2) Quantitative Evaluation Metrics: We evaluated the output of our TDPT pose estimation pipeline both quantitatively and qualitatively. We compared TDPT along with our original approach, OpenPose+MotioNet, against ground truth CMU motion capture data using two key metrics: temporal inconsistency and average limb length variation.

Temporal inconsistency measures the frame-to-frame choppiness of joint motion, and a low score reflects stable tracking. Limb length variation quantifies how much each limb's length changes over time as a function of the mean and standard deviation of limb lengths. Again low scores indicatie more realistic reconstructions, as limbs shouldn't change in size. Formulas for each are given below:

$$\text{Temporal Inconsistency} = \frac{1}{N}\sum_{t=1}^{N-1}\frac{1}{j}\sum_{j=1}^{J}||p_{t+1,j} - p_{t,j}||$$

$$\text{Avg Limb Length Variation} = \frac{\text{Std}(\ell_{a,b})}{\text{Mean}(\ell_{a,b})} \times 100$$

B. Motion Retargeting

1) Data Preprocessing: We represent a motion sequence as a combination of a static component $S \in \mathbb{R}^{J\times S}$ and a dynamic component $Q \in \mathbb{R}^{T\times J\times Q}$, where J is the number of armatures and T is the sequence length.

The static component S consists of a set of offsets in 3D vectors ($S = 3$), describing the properties of each skeleton. The dynamic component Q consists of a set of unit quaternions ($Q = 4$), representing the rotation information of each joints (relative to its parents' coordinate frame).

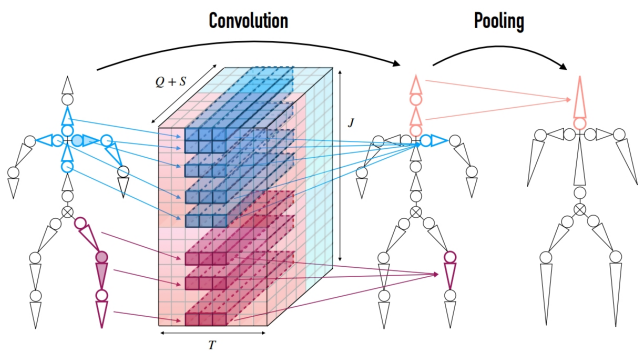


Fig. 3: The illustration of convolution and pooling operators[18]. The unpooling operator is the counterpart of pooling.

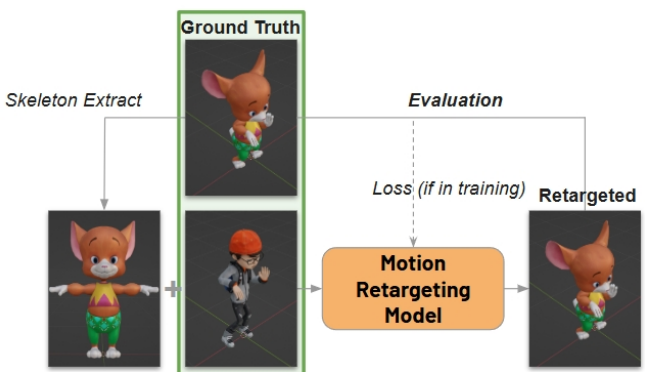


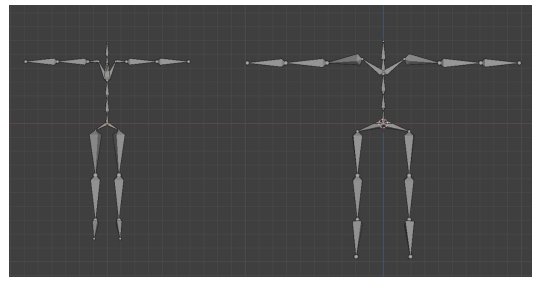
Fig. 4: Training and Evaluation Process. Metrics are calculated between retargeted motion and ground truth motion on same character. There is no joint matching issue even in cross-structural retargeting scenario.

To obtain adjacency information, the skeleton structure is treated as a tree graph whose nodes correspond to joints and edges correspond to armatures. Then, the connectivity information can be inferred through adjacency lists $N^d = \{N_1^d, N_2^d, ..., N_J^d\}$, where N_i^d contains edges whose distance in the tree is equal or less than d from the i-th edge.

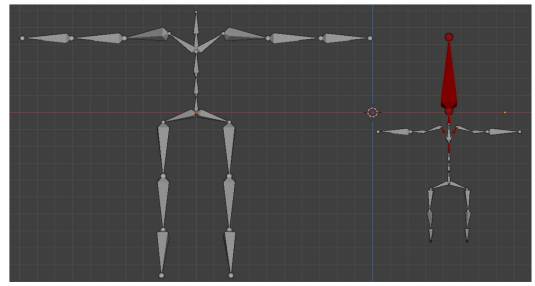
2) Skeleton-Aware Network(SAN) Reproduction: Some mainstream pose estimation methods typically simplify the human body into a skeleton and output the corresponding information. Therefore, using SAN allows us to better adapt to this type of output. Additionally, SAN has a good performance while maintaining a moderate level of complexity, achieving a good balance between model performance and efficiency.

SAN consists of three main components (Figure 3):

- **Convolution Layer**: Applies skeleton-aware convolution considering the dynamic and static information for local feature extraction.
- **Pooling Operator**: Reduced the skeletons to a primal skeleton by sequentially merging pairs of adjacent edges to extract shared latent domain-space features.
- **Unpooling Operator**: Expand the number of edges by copying the feature activations of each edge that is originated by a merging of two edges in the corresponding pooling step.



(a) Intra-Structural Difference



(b) Cross-Structural Difference

Fig. 5: Skeleton Differences. The red parts represents extra joints.

These three operators form the core of the skeleton-aware motion processing framework, enabling efficient, accurate, and smooth motion retargeting across different skeletons.

We first trained SAN on the Mixamo dataset (Figure 4), which contains hundreds of characters and thousands of motions. These motions have been retargeted and refined by professional animators, so we can use them for our supervise learning. After training, we directly integrate SAN into our pipeline without further tuning.

3) Quantitative Evaluation Metrics: SAN is capable of handling different skeleton structures, which are also mixed together during training to improve generalization ability. However, in the evaluation phase, we categorize these structure differences into two major types (Figure 5):

- **Intra-Structural Differences**: Skeletons has the same topological structure but different proportions.
- **Cross-Structural Differences**: Skeletons has a different topological structure, such as extra leg/arm joints.

We can then perform a more accurate and specific evaluation within each category. We evaluate retargeting efficiency and accuracy using Blender Python Scripts and process in Figure 4. The formulas are as follows:

$$\text{efficiency} = \frac{\text{total frames produced}}{\text{total time consumed}}$$

$$\text{error} = \text{local joints error} = \sum_{i}^{N} \sum_{j,t}^{J_i,T_i} \left(\frac{||p_{i,j,t} - \hat{p}_{i,j,t}||_2}{h_i} \right) / N$$

N represents the number of motion sequences in the test set. J_i is the number of joints for i-th sample's character, and T_i is the number of frames for i-th sample's motion. $p_{i,j,t}$ is the true position(world frame) of the j-th joint in i-th motion sequnce at t-th frame, while $\hat{p}$ indicates the retargeted

position. h_i refers to the height of the i-th sample's character, defined as the z-value of the highest joint node.

C. Whole Pipeline

We then construct our whole pipeline by integrating TDPT and SAN, and test it on various real human videos.

1) Test Data Preparation: We utilize data from the CMU Panoptic Dataset, segmenting it into several clips that highlight distinct body movements. The dataset provides high-resolution video inputs along with corresponding 3D keypoint ground truth, making it well-suited for both pose estimation and quantitative evaluation.

2) Blender Visualization: We convert the retargeted motion data into a BVH file, which can be directly loaded for animation visualization in Blender. Using Blender's Animation Editor, we can intuitively compare motion differences in a sequential manner(animation). Through Blender's Python Script API, we can perform a quantitative evaluation of motion retargeting performance.

3) Quantitative Evaluation Metrics:

Strict self-penetration ratio: For each frame f let J be the total number of mesh vertices and P_f the number of vertices that belong to **any** pair of triangles that truly intersect (exact triangle–triangle test, no tolerance). The per-frame ratio is

$$\rho_f = \frac{P_f}{J} \quad (0 \le \rho_f \le 1).$$

From this we report

$$\text{max pen.} = 100 \times \max_f \rho_f, \quad \text{avg pen.} = 100 \times \frac{1}{N} \sum_{f=1}^{N} \rho_f,$$

and the percentage of frames where $\rho_f > 0$.

Smoothness score (0–200): For every bone b let $\theta_b(f)$ be its rotation angle in radians at frame f. Using forward differences

$$v_b = \Delta\theta_b,\ a_b = \Delta v_b,\ j_b = \Delta a_b,$$

the bone's root-mean-square jerk is

$$J_b = \sqrt{\frac{1}{N-3} \sum_{f=1}^{N-3} \big(j_b(f)\big)^2}.$$

Averaging over all B bones gives $J = \frac{1}{B}\sum_b J_b$. With a tunable threshold $\texttt{SCALE} = 0.05$ rad frame^{-3} we map to

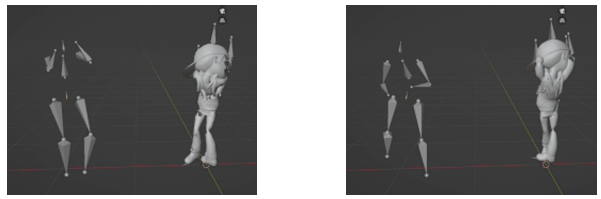
$$\text{Score} = 200\ \max\Big(0,\ 1 - \tfrac{J}{\text{SCALE}}\Big),$$

so perfectly smooth motion scores 200 while $J \ge \text{SCALE}$ scores 0.

IV. Results

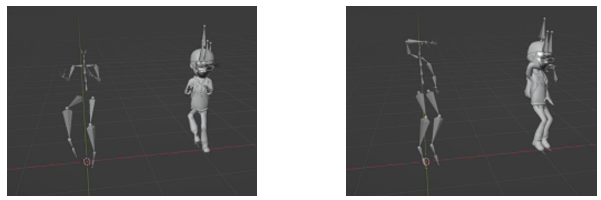
A. Pose Estimation

We evaluated the output of our TDPT pose estimation pipeline both quantitatively and qualitatively. We compared TDPT along with our original approach, OpenPose+MotioNet, against ground truth CMU motion capture data using two key metrics: temporal inconsistency and average limb length variation.



(a) (b)

Fig. 6: Retargeting result: BVH file generated by MotioNet



(a) (b)

Fig. 7: Retargeting result: BVH file generated by TDPT

TDPT achieved the best results, with a temporal inconsistency of 0.0468 and limb length variation of 5.71%, closely matching ground truth (0.0672, 2.32%) and substantially outperforming OpenPose+MotioNet (0.688, 8.79%). This indicates that TDPT produces much smoother and more anatomically stable motion, while OpenPose+MotioNet is prone to choppy motion and unrealistic limb length fluctuations. These results are summarized in Table I

TABLE I: Comparison of Pose Estimation Performance Metrics (lower = better)

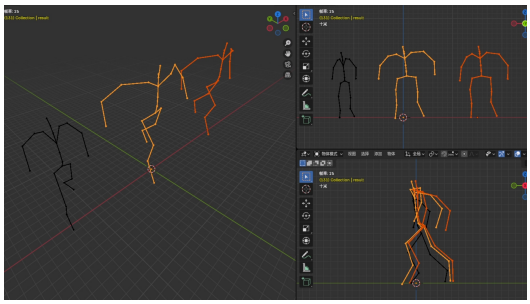
	Temporal Inconsistency	Limb Length Variation
Ground Truth (CMU)	0.0672	2.32%
TDPT	0.0468	5.71%
OpenPose+MotioNet	0.688	8.79%

Qualitative results further support these findings. From Blender visualizations of our reconstruction images, one can see that TDPT's 3D output preserves natural body orientation and smooth joint rotations, resulting in visually plausible retargeted motion. This is demonstrated with a freezeframe in Figure 7 In contrast, MotioNet-based reconstructions exhibited severe artifacts such as torso flipping and incorrect limb directions, confirming that small pose estimation errors (especially in 3D joint rotations) can propagate and cause major failures in retargeting. This is shown in Figure 6

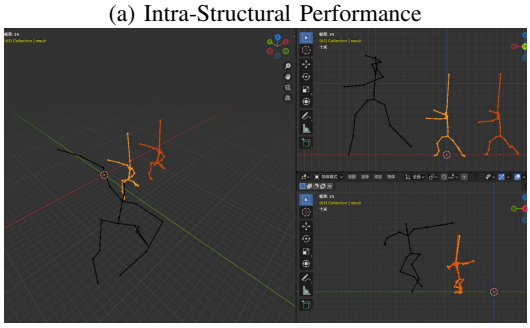
In summary, TDPT demonstrated superior quantitative and qualitative performance, making it the preferred choice for our pipeline.

B. Motion Retargeting:

1) Visual Comparison on Different Setups: We test SAN on two different setups: Intra-Structural Retargeting, Cross-Structural Retargeting (Figure 8). By comparing the anima-



(a) Intra-Structural Performance



(b) Cross-Structural Performance

Fig. 8: Motion Retargeting Comparison. We extract the frame at the middle of a motion sequence with three perspective. The black skeleton represents the input source motion. The yellow skeleton represents the retargeted motion. The orange skeleton represents the ground truth motion.

tions in Blender, we can confirm that SAN achieves decent reconstruction accuracy and smoothness.

2) Quantitative Evaluation: Quantitative evaluation is conducted based on Mixamo test dataset.

$$\text{efficiency} = 147 \text{ frames/second (without rendering)}$$

$$\text{error} = 0.87\% \text{ (intra-structural)}$$

$$\text{error} = 2.54\% \text{ (cross-structural)}$$

These results demonstrate that SAN achieves good efficiency while maintaining low positional error. For example, assuming a standard avatar height of 1.5 meters, the average cross-structure retargeting error is approximately 4 cm. This level of error is acceptable and does not significantly affect motion expression.

C. Whole Pipeline

We conduct experiments on real-world human videos to assess the performance of our pipeline. For a better comparison, we implemented Auto Rig Pro (ARP) as a baseline for the motion retargeting. ARP is a Blender Add-on that handles motion retargeting problems in a traditional way. It involves a series of bone matching, Inverse Kinematics, and Forward Kinematics rigs.

1) Visual Comparison: According to the visualization result (Figure 9), we can observe that:

- SAN can retarget most motions.
- SAN works well on both intra-structure and cross-structure motion retargeting.
- SAN outperforms ARP in terms of penetration.

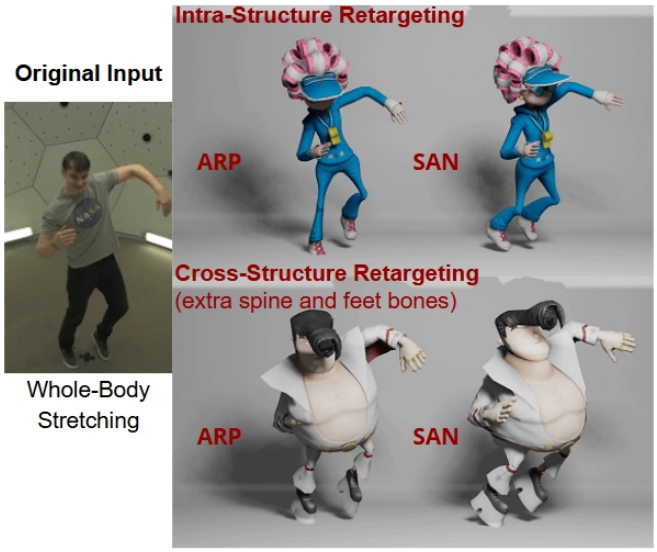


Fig. 9: Pipeline Results.

2) Penetration Evaluation: As seen in table II, across the ten motion sequences SAN consistently lowers the proportion of vertices that intersect the mesh when compared with the ARP baseline. The average penetrating fraction decreases from 50.24 These results show that SAN preserves surface integrity more effectively than ARP.

3) Smoothness Evaluation: SAN also produces smoother joint trajectories. The mean 0–200 smoothness score rises from 149.49 to 155.48 for intra-structure pairs and from 139.00 to 146.73 for cross-structure pairs, indicating a lower global RMS jerk in both cases.

Gains are most noticeable on highly articulated motions: the lateral-arm raise improves by 27 points, and the front-arm raise improves by 6 points. The smoothness of SAN produced motion is further validated by figure 11, where more jerky movement is seen in the blue curve representing the motion of ARP.

Overall, SAN delivers both cleaner surfaces and less jittery motion than the ARP baseline.

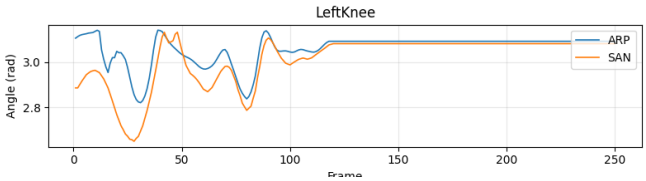


Fig. 10: Comparison motion plot of left knee movement

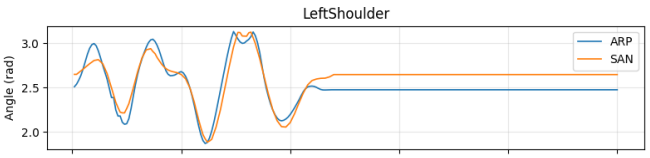


Fig. 11: Comparison motion plot of left shoulder movement

TABLE II: Penetration Evaluation across Ten Motion Sequences

Primary Body Part	**Intra-Structure Retargeting**		**Cross-Structure Retargeting**	
(Motion Type)	ARP penetration(%)	SAN penetration(%)	ARP penetration(%)	SAN penetration(%)
arm (lateral raise)	52	51	31.51	27.12
arm (front raise)	51.2	49.6	31.6	28.5
neck (twist)	51.4	48.4	33.7	28.4
shoulder (shrug)	51.1	47.97	32.7	30
waist (twist)	47.8	47.1	32	30.3
elbow (twist)	49.1	48.3	27.17	25.49
leg (lift)	49.18	47.85	29.7	28
back (bend over)	50.41	50.32	33.66	33.34
leg (walk/trot)	50.93	49	32.85	29.74
whole-body (stretch)	49.27	**47.6**	29.91	**27.64**
average	**50.24**	**48.71**	**31.48**	**28.85**
avatar (before retargeting)	**51.06**		**26.35**	

TABLE III: Smoothness Evaluation across Ten Motion Sequences

Primary Body Part	**Intra-Structure Retargeting**		**Cross-Structure Retargeting**	
(Motion Type)	ARP smoothness score	SAN smoothness score	ARP smoothness score	SAN smoothness score
arm (lateral raise)	140.5	167.4	126.2	151.2
arm (front raise)	151	167.2	142.6	157.7
neck (twist)	191.8	192.3	193.1	192.5
shoulder (shrug)	179.2	178.6	172.8	173.7
waist (twist)	154.4	158.9	155.9	154.9
elbow (twist)	144.1	138.1	123	119.3
leg (lift)	160	153.3	158	151.8
back (bend over)	93.2	**–**	62.2	**–**
leg (walk/trot)	131.2	139.1	117.2	132.7
whole-body (stretch)	–	104.4	–	86.8
average	**149.49**	**155.48**	**139.00**	**146.73**

V. Discussions

A. Pose Estimation

Our investigations into pose estimation in this project demonstrates the impact of pose accuracy, particularly with respect 3D joint rotations, on the quality of downstream motion retargeting. While our initial pipeline leveraged OpenPose for 2D keypoint extraction and MotioNet for 3D motion reconstruction, we observed that inferring joint rotations from 2D data alone resulted in severe errors when retargeted to new skeletons. In contrast, TDPT produced significantly more stable results, as evidenced by both qualitative animation (see Fig. 7) and improved consistency metrics (see Table I).

We infer the key reason behind MotionNet's poor performance lies in the difficulty of inferring 3D joint rotations from only 2D keypoints. Since 2D poses lack depth and contain inherent ambiguity, it is a big challenge to recover accurate 3D orientations. In addition, MotionNet may not apply sufficient constraints to ensure physically plausible joint movements, which can result in discontinuities or even flipping.

Meanwhile, TDPT can use more robust video-based cues, such as optical flow, 3D body mesh estimation, or learned kinematic priors, which allow it to recover global rotations and joint orientations more accurately.

Although TDPT is a closed-source, black-box tool, its robust rotation estimation and BVH export made it the preferred choice for our current pipeline. Its superior performance acts as motivation to integrate temporal modeling and biomechanical constraints into open frameworks to achieve both accuracy and adaptability in pose estimation.

B. Motion Retargeting

Based on our evaluation, we find that SAN shows good performance with Mixamo dataset and can serve as a key component of our pipeline. The results on real human videos indicate that SAN handles most human motions effectively in terms of smoothness and penetration. However, its robustness degrades when applied to real human skeletons and motion data. There could be three potential reasons for this.

First, SAN ideally needs to be trained on new skeleton structures to achieve best performance. However, we are unable to do fine-tuning on a human skeleton dataset since the ground truth motion is missing. To help improve adaptability and accuracy, we need the pose estimation output to be similar to the Mixamo skeleton structure.

Second, human motions are typically more intense and unpredictable. SAN struggles to handle such motions. A thorough evaluation of SAN across diverse motion types is needed to identify its specific limitations and define its applicability boundaries for real-world scenarios.

Third, SAN incorporates temporal information, which means that a wrong pose estimation frame—such as incorrect foot orientation caused by occlusion—can propagate through subsequent frames and lead to cascading failures. In such

cases, the performance of SAN heavily depends on the quality of the pose estimation results. One potential approach to enhance its robustness is to improve pose estimation accuracy using stereo or monocular depth cameras.

C. Future Work

Moving forward, we plan to enhance the robustness of SAN by improving its tolerance to ill-posed or noisy input poses, ensuring more stable retargeting outcomes. To streamline the entire pipeline, we aim to develop an end-to-end automation framework that enables seamless data flow between modules without manual intervention, ultimately improving usability and integration. We also intend to systematically investigate the effects of camera constraints—such as resolution, perspective, and distance—on retargeting performance, which may reveal important insights for broader deployment. Lastly, we will explore reproducing the TDPT pose estimation method to better understand its internal mechanisms and potentially improve the accuracy and consistency of pose inputs, enabling tighter coupling between perception and retargeting components.

VI. Conclusion

In this study, we investigated a pose-estimation and motion-retargeting framework initially combining OpenPose and MotionNet with a Skeletal Aware Network (SAN). Our experiments indicated that the `.bvh` outputs from OpenPose and MotionNet suffer from problematic link rotations, rendering them incompatible with SAN's retargeting algorithm. This incompatibility was further validated by difficulties in successfully attaching skin or meshes to the resulting `.bvh` files. By contrast, we discovered that the open-source TDPT tool generates more accurate rotational data, as verified by successful skin attachments to its `.bvh` outputs. Consequently, TDPT presents itself as a more robust option for pose estimation in this pipeline. Additionally, we found that SAN's pre-trained model does not generalize well across diverse skeletal structures, hinting at the necessity for fine-tuning on custom skeletons. Lastly, laveraging ARP - which utilize traditional kinematic calculations - helped mitigate retargeting artifacts and provided more stable motion transfer, making it a promising addition to the framework.

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